Jeongah Jasmine Lee

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jeongahlee@umass.edu Personal Website Linkedin Google Scholar

Research Interests

Human-Computer Interaction, Human-AI Interaction, Visualization, Accessibility

Education

MS/PhD **University of Massachusetts Amherst**, Computer Science — Sep 2024 - Present
BS **Sungkyunkwan University**, Computer Science and Engineering — Mar 2019 - Feb 2024
The University of Texas at Austin (Exchange Student Program) — Jan - May 2022

Publications

[P.4] En3: A Tripartite Approach Toward Holistic Understanding and Detection of Engagement in Educational Video Content
- Sidharth Anupkrishnan, Itir Sayar, **Jeongah Lee**, Sri Harsha Musunuri, Guan-Ming Su, Tam Vu, Madeline Endres, Ravi Karkar, Phuc Nguyen
- Manuscript in Preparation

[P.3] Resolving Ruptures: Mental Health Professionals' Perspectives on How LLMs Identify and Resolve Conversational Ruptures
- ***Jeongah Lee***, Qiuyue Joy Zhong, Drishti Goel, Violeta J. Rodriguez, Dong Whi Yoo, Koustuv Saha, Ravi Karkar
- Under Review

[P.2] Conversational AI for Floorplan Accessibility: How Blind and Low-Vision Users Explore Space Through Dialogue
- ***Jeongah Lee***, Srikiran Kavuri, Hari Iyer, Ali Sarvghad
- Under Review

[P.1] Cohort-based Semantic Labeling: AI-Enabled Recovery of Visualization Semantics from Deployed SVGs
- ***Jeongah Lee***, Hima Varshini Surisetty, Durga Nirmaleswaran, Jahnavi Sharma, Srikiran Kavuri, Narges Mahyar, Ali Sarvghad
- Under Review

[C.2] RubRIX: Rubric-Driven Risk Mitigation in Caregiver-AI Interactions
- Drishti Goel, ***Jeongah Lee***, Qiuyue Joy Zhong, Violeta J. Rodriguez, Daniel S. Brown Ravi Karkar, Dong Whi Yoo, Koustuv Saha
- 2026 Findings of the Association for Computational Linguistics (ACL) (Acceptance rate: 18%)

[C.1] From Text to Visuals: Using LLMs to Generate Math Diagrams with Vector Graphics
- Jaewook Lee, ***Jeongah Lee***, Wanyong Feng, Andrew Lan
- AIED 2025: 26th International Conference on Artificial Intelligence in Education (Acceptance rate: 19%)

[J.1] IoT Edge-Cloud: An Internet-of-Things Edge-Empowered Cloud System for Device Management in Smart Spaces
- Yoseop Joseph Ahn, Minje Kim, ***Jeongah Lee***, Yiwen Shen, Jaehoon Paul Jeong
- 2023 IEEE Network Magazine

Research Experiences

Dolby Advanced Technology Group, PhD Research Intern — ***Atlanta, GA, USA*** — May - Aug 2026
- Advisor: Dr. Gautam Bhattacharya

HCI-VIS lab, WHAT lab @ University of Massachusetts Amherst, Graduate Research Assistant — ***Amherst, MA, USA*** — Sep 2024 - Present
- Advisor: Dr. Ravi Karkar, Dr. Ali Sarvghad
- Led the design and implementation of human-centered AI systems for visualization **[P.1]**, accessibility **[P.2]**, and health **[P.3]**, spanning user research, system prototyping, and evaluation
- Led research on AI-driven mental health tools for Alzheimer's caregivers, integrating journaling, conversational interaction, and cognitive reframing (CBT) support **[C.2, P.3]**

IoT lab @ Sungkyunkwan University, Undergraduate Student Researcher — ***Suwon, Korea*** — Jul - Nov 2021
- Advisor: Dr. Jaehoon Paul Jeong
- Developed an IoT edge-empowered Cloud System for the visual control of IoT devices in a user's smartphone **[J.1]**

Work Experiences

Seoul National University Bundang Hospital, Natural Language Processing (NLP) Researcher — ***Bundang, Korea*** — Mar - Jul 2024
- Developed a model predicting lung cancer TNM stage using an Electronic Health Record dataset
- Fine-tuned Large Language Models (LLMs) in resource-restricted settings, optimizing model performance through tailored prompt engineering techniques

Cipherome, Inc, Machine Learning (ML) Engineer, Intern — ***San Jose, CA, USA*** — Mar - Aug 2023

- Developed the pipeline for an ML module within a clinician-focused medical data analysis platform
- Designed Figma wireframes to improve the platform's user interface and experience (UI/UX)

SK Planet Co.,Ltd., Industry-Academic Cooperation Student Researcher — ***Suwon, Korea*** — Mar - Dec 2020

- Developed an AR rhythm game application for the Busan One Asia Festival using AR Core to provide location-based interactive experiences

Awards & Honors

[Awards]

- 3rd Place (Grand Prize), Chung-ang University AI and Humanities Academic Paper contest — 2023
- 1st Place (Grand Prize), Kookmin University self-driving contest — 2021
- 3rd Place (Grand Prize), Sungkyunkwan University AI x Bookathon contest — 2021
- Volunteering Excellence Prize, NIA(National Information Society Agency) — 2020

[Scholarships]

- Academic Excellence Scholarship (top 12%) — 2022
- Creative Scholarship (100% tuition support) — 2021
- Sungkyun Software Scholarship (100% tuition support) — 2019
- MegastudyEdu Scholarship (external) — 2019

Teaching & Mentoring

[Teaching]

Graduate Teaching Assistant — Sep 2024 - Dec 2025

- *CICS 110 Introduction to Programming (Fall 2024)*: Led two lab sessions (60 students each) with hands-on instruction and grading support.
- *CS 383 Artificial Intelligence (Spring 2025)*: Mentored students in 15 group projects and assisted with assignment design and evaluation (180 students).
- *CS 571 Data Visualization and Exploration (Summer 2025), INFO 348 Data Analytics with Python (Fall 2025)*: Sole TA for CS 571 (35 students) and INFO 348 (60 students); independently managed grading, office hours, and assignment design.

[Mentoring]

Undergraduate Research Volunteer Program Mentor — May - Sep 2025

- Gerindra Adi, Vidhaan Kothari, Nish Methuku (CS Undergraduate)
- Topic: Glanceable Health Visualization for Older Adults Using Large Multimodal Models

Services

IEEE VIS Poster Committee — 2026
Eurovis Reviewer — 2026

Volunteering

Panama World Friends Korea ICT e-volunteer — Nov - Dec 2020
Mobile Application Programming Mentoring at Youngbok Girls' High School — Sep - Dec 2019
College of Computing Student Council — Mar - Dec 2019

Skills

Technical Proficiencies: Python, Pytorch, JavaScript (React.js, D3.js, Node.js), Flask, HTML, CSS, MongoDB
Research Methodologies:
Qualitative: User Research, Semi-Structured Interviews, Prototyping, Usability Testing, Think-Aloud Protocol, Thematic Analysis
Quantitative: Survey Design, Experimental Design, Statistical Analysis, Modeling, Visualization Analysis
Tools & Platforms: Git, GitHub, Vercel, Render, Capacitor, Docker, Figma, Zeplin